

MICROVISION

System Changes & Additions

Living Document v1.3 — New changes added here as discussed. Each change is self-contained.

Change 1: Skip SC + Search · Change 2: Unregistered SC · Change 3: New Step 2 + Warranty ·
Change 4: Billing Filters + Paid System

Change 1 — Skip SC Assignment + Advanced Complaint Search

Two related improvements to the complaint registration and search experience.

1A. Skip SC Assignment on Step 4

Step 4 of the complaint registration form previously required admin to assign a Service Centre before submitting. Assignment is now optional.

What changes

- Step 4 now has two paths: Assign SC (existing flow, unchanged) OR Skip — Assign Later.
- If admin clicks Skip: complaint is created with status unassigned. No SC is linked. No WhatsApp is sent.
- The complaint appears in All Complaints with a clear Unassigned badge.
- Admin opens the complaint later and assigns an SC from the detail view — same SC picker, same capability filtering as Step 4.
- Once assigned from detail view: status moves to assigned, WA-01 fires to SC immediately. Full normal flow resumes.

Status addition

- unassigned is a new valid status before assigned in the lifecycle. Complaint exists but has no SC yet.
- Admin Action Centre shows unassigned complaints as a dedicated section with an Assign button.
- No WhatsApp, no automated reminders for unassigned complaints. Admin handles manually.

unassigned → assigned happens when admin assigns from detail view. From assigned onwards the flow is identical to normal.

1B. Advanced Complaint Search — Unified Top Bar + Multi-Filter

Top Search Bar — 5 Fields Simultaneously

The main search input searches all 5 fields at once with a single query. Admin types anything — system checks all 5 and returns matches:

Field	Matches Against
Customer Name	customerName (partial, case-insensitive)
Phone Number	phone1 OR phone2 (partial match)
Complaint ID	complaintId (partial match)
Serial Number	serialNumber on linked Product record
Product ID	productId on linked Product record

- Results update as admin types (debounced — fires after 300ms pause).

- No need to know which field the value belongs to — just type and search.

Expanded Filters — All Combinable Simultaneously

All filters work together (AND logic — results must match every active filter). Filters available:

Filter	Type	Options
Status	Multi-select	unassigned / assigned / accepted / going / done / not_done / part_pending / closed / rejected_by_sc. Multiple selectable at once.
Product	Single select	All / LED / Cooler
Complaint Type	Single select	All / Installation / Complaint
Warranty	Single select	All / In Warranty / Out of Warranty
State → District → City	Cascading selects	State filters districts, district filters cities
Assigned SC	Single select	All registered + unregistered SCs
SC Capability	Single select	All / LED Only / Cooler Only / Both
Tracking ID	Text input	Partial PT-XXXXXX match
Serial Number	Text input	Partial serial match
Reopen Status	Single select	All / Reopened Only / Original Only
Date Range	Date pickers + shortcuts	Today / Yesterday / Last 7 days / Last 30 days / Custom From→To

- Reset button clears all filters and search at once.
- Active filter count shown on the Hide/Show Filters toggle button.
- Example: search '9876' in top bar + Status = done + District = Jodhpur + Date = Yesterday — returns all done complaints with that phone fragment in Jodhpur registered yesterday.

Change 2 — Unregistered (Admin-Maintained) SC

Sometimes a complaint comes from a location where no registered SC exists. Admin can now create and assign an Unregistered SC on the spot, maintain it fully, and optionally convert it to a full registered account later. All history is preserved throughout.

2A. Creating an Unregistered SC

During Step 5 (previously Step 4) or when assigning from complaint detail, a new option appears alongside the normal SC list:

+ Assign to New Unregistered SC (not in system)

Admin fills a minimal form:

Field	Required?
Name / Business Name	Yes
Phone Number 1	Yes
Phone Number 2	No
City	Yes — from master city list (or admin creates new city on the spot — see Section 2D)
District	Auto-filled from city
Full Address	No

- A new SC record is created with flag: isUnregistered: true.
- Gets a system-assigned SC ID just like a normal SC.
- Complaint is immediately assigned to this unregistered SC.
- UNREGISTERED SC badge appears everywhere this SC or their complaints are shown.
- No login credentials created. Admin contacts them manually outside the app.

2B. WhatsApp Behaviour for Unregistered SC Complaints

- WA-04 to customer STILL fires immediately on assignment — customer deserves to know their complaint is assigned and who to contact.
- Content of WA-04: Complaint ID, Product type, Complaint type, SC name, SC phone number, acknowledgment message. Identical to registered SC assignment.
- ALL other WhatsApp messages (SC reminders WA-01/02/03, part delivery WA-06, post-received WA-07) do NOT fire. Admin handles SC communication manually outside the app.

Customer sees no difference between a registered and unregistered SC assignment. The distinction is purely internal.

2C. Admin Maintains All Updates for Unregistered SC Complaints

- Steps 1, 2, 3 of complaint registration are identical — customer info, product info, charges, media.

- After assignment to unregistered SC, admin manually calls/contacts them in real life.
- Admin updates all complaint statuses on the SC's behalf: Going, Done, Not Done, Part Pending.
- Admin fills all related forms (Done form: photos, notes, petrol, extras) based on information received from the unregistered SC.
- Product tracking, complaint timeline, billing logic all work identically to normal complaints.
- No automated reminder cycles fire. Admin is solely responsible for follow-up.

The UNREGISTERED SC badge on the complaint signals to admin that manual follow-up is required at every stage.

2D. New State, District, or City — Admin Can Create on the Spot

If an unregistered SC (or any SC registration) is from a location not in the master city list, admin can create it immediately from any city/district/state field in the system:

- New State — if the state does not exist yet (e.g. Haryana, Gujarat before formal seeding).
- New District — if the state exists but the district does not.
- New City — if state and district exist but the city does not.
- When admin types an unlisted name in a city field: a 'Create new' option appears inline.
- Admin fills: city name, district name, state name — saved to cities collection immediately.
- New city/district/state is instantly available across ALL dropdowns, filters, and search fields system-wide.
- This works from: Unregistered SC creation form, SC registration approval screen, and anywhere else admin enters a city.

2E. Unregistered SCs Are Searchable and Reusable

- All unregistered SC records appear in the Service Centres tab with UNREGISTERED badge.
- Filterable: admin can filter SC list to show Unregistered only.
- Searchable by: name, phone, city, district — same as registered SCs.
- When assigning a future complaint in the same area, admin can select an existing unregistered SC from the list instead of creating a new one.
- Full complaint history and billing history maintained under each unregistered SC record.

2F. Upgrading Unregistered SC to Registered SC

When an unregistered SC decides to formally join the platform:

Step 1 — SC submits normal registration

- They use the public registration page and fill the full SC form as normal.
- Their request appears in admin Action Centre as a pending SC registration.

Step 2 — Admin links during approval (search-based, not phone-match only)

- System checks for phone number matches as a suggestion — hint only, not required.

- Even if phone numbers do not match, admin can manually search all unregistered SC records by name, city, district, or address and pick the correct one.
- Admin confirms the link manually. No phone match required — admin's judgment is sufficient.

Step 3 — On linking: all history preserved exactly as-is

- All complaints, timeline entries, admin-filled forms, photos, notes, billing records, product tracking links transfer to the new registered SC account.
- Old UNREGISTERED SC badges and admin-maintained markers remain in complaint timeline as audit history — clear record of who did what and when.
- isUnregistered flag removed. UNREGISTERED badge disappears from all active views.
- SC now follows full normal registered flow: portal access, WhatsApp notifications, SC-side form submissions, reminder cycles.
- Old unregistered SC record is archived (not deleted) for audit purposes.

If admin dismisses the link and approves without linking: a separate registered SC account is created with no connection to old unregistered record. Old record and history remain as-is.

Change 3 — New Step 2: Product Info + Warranty Overhaul

A new Step 2 is inserted into the complaint registration form. Current Step 2 (Product & Type), Step 3 (Charges & Media), and Step 4 (SC Assignment) shift to Step 3, Step 4, and Step 5 respectively. Step 1 (Customer Info) is unchanged.

New complaint form step order: Step 1 — Customer Info | Step 2 — Product Info (NEW) | Step 3 — Product & Type | Step 4 — Charges & Media | Step 5 — SC Assignment

3A. New Step 2 — Product Info Fields

All 5 fields are optional but recommended. Admin should fill as many as possible for long-term product tracking and warranty accuracy.

Field	Type	Optional?	Visible to SC?	Notes
Bill Date	Date picker	Optional (recommended)	No — admin only	Used for automatic warranty calculation (bill date + 3 years). If filled, warranty is auto-calculated. Saved to Product record.
Bill Photo	Image upload	Optional (recommended)	No — admin only	Photo of original purchase bill/receipt. Stored in product's bill photo field. If SC uploads later, auto-fills here.
Shop Name	Free text	Optional (recommended)	No — admin only	Name of shop/dealer where product was purchased. Free text, any value.
Serial Number	Free text	Optional (recommended)	Yes — sent to SC in WA-01 if filled	Unique identifier for the physical unit. Saved to Product record. Used for product tracking and lookup.
Model Number	Free text	Optional (recommended)	Yes — sent to SC in WA-01 if filled	Model/variant of the product. Free text. Saved to Product record.

These 5 fields are saved to the Product record (product tracking layer), not just to this complaint. Any value filled here persists and pre-fills for all future complaints linked to the same product.

3B. Pre-fill Behaviour When Product is Already Linked

- If Step 1 finds a linked product (by phone or serial match): Step 2 comes pre-filled with all 5 fields from the Product record.
- Admin can edit any pre-filled value. If admin changes a value that was previously saved: a warning appears inline — 'This field was previously saved as [X]. You are changing it to [Y].' Admin confirms or reverts.
- On complaint submission, updated values are saved back to the Product record — self-correcting over time.

3C. Location / Action Text Field (Step 1 Addition)

A small optional field is added to Step 1 (Customer Info):

Field	Type	Required?	Purpose
Location / Action Text	Free text (paste-friendly)	Optional	If customer shares a Google Maps link, coordinates, or any location description, admin pastes it here. Sent to SC in WA-01 message so SC can navigate directly.

- No validation — any text accepted (Maps link, address description, landmark, coordinates).
- Visible to SC in their complaint detail view and in WA-01 message.
- Stored on the Complaint record — not on the Product record (location may differ per visit).

3D. SC Done Form — Demanded Fields Based on What's Missing in Step 2

When SC submits their Done form, the system checks which Step 2 fields were filled at registration. Missing fields trigger demands in the Done form:

What's Missing in Step 2	What SC Done Form Demands	Required?
Any of: Bill Date / Bill Photo / Shop Name	Bill receipt / photo upload from customer — SC must obtain and upload the bill	Warned but bypassable
Serial Number OR Model Number (or both)	Photo of the product's serial number / model sticker (physical slip on the product)	Warned but bypassable

- These demanded uploads appear as a dedicated sub-section in the SC's Done form, clearly separated from proof photos.
- If SC does not upload the demanded items and clicks Submit: a warning is shown listing what is missing. SC must either go back and upload OR explicitly bypass each item.
- Bypass is allowed — SC selects 'Skip for now' per item. System records that it was skipped (not uploaded by SC).

3E. SC Upload — Bill Photo and Serial Slip Handling

- Bill photo uploaded by SC in the Done form: automatically saved to the Product record's bill photo field. Admin can see it from the complaint's product info subsection, and can keep it, remove it, or replace it with a different photo. If SC uploads, it becomes the active bill photo on the Product record immediately.
- Serial number slip photo uploaded by SC: stored as a separate media item on the complaint (not in the Product record's main fields). Admin views it in the complaint's product info subsection and can manually transcribe the serial/model number into the Product record fields.

- Both uploads are NOT visible to SC after submission — they go into admin's product info view only.

3F. Admin at Closing Time — Final Field Check

When admin is about to confirm Done and close the complaint, the system checks all 5 Step 2 fields on the Product record. Any that are still empty trigger a warning before closing:

- Warning shown: 'The following product fields are still empty: [list of missing fields]. Please fill them or bypass to close.'
- Admin can go back to the product info section, fill the missing fields (using SC-uploaded bill photo or serial slip if SC provided them), and then close.
- Admin can bypass each warning individually — explicit bypass required per field, not a single 'skip all'.
- If any field is still missing after bypass at closing: a permanent indicator remains on the Product record showing which fields were never filled. This indicator shows as a warning on all future complaints for the same product until the fields are eventually filled.

This persistent warning ensures that every future admin handling a complaint for this product is reminded to fill the missing fields, even if previous admins bypassed them.

3G. Warranty Logic — Final Consolidated Rules

Warranty status is determined at every complaint creation and stored as a snapshot on both the Complaint record and the Product record. The following rules apply in priority order:

Rule 1 — Bill Date present

- `warrantyExpiryDate` = `billDate` + 3 years
- `warrantyStatus` = In Warranty if `today` ≤ `warrantyExpiryDate`, else Out of Warranty
- Shown to admin as: 'In Warranty (expires DD MMM YYYY)' or 'Out of Warranty (expired DD MMM YYYY)'
- `warrantySource` = `auto_calculated`

Rule 2 — No Bill Date

- Admin must manually select: In Warranty or Out of Warranty
- Admin must provide a reason for the manual selection (free text, required)
- `warrantySource` = `manual`
- Reason is stored on the Product record and visible to admin only — SC sees only 'In Warranty' or 'Out of Warranty', never the reason

Rule 3 — LED Installation (special default)

- LED Installation complaints are always In Warranty by default — a brand new install is obviously in warranty
- If admin fills bill date: warranty is auto-calculated from that date for all future complaints on this product

- If admin does not fill bill date: warrantyStatus = In Warranty, warrantySource = manual (default), no expiry date stored
- Force override always available even for installations — admin can force Out of Warranty with a reason in rare edge cases

Rule 4 — Force Override (always available)

- Even if bill date is present and auto-calculates a status, admin can always force override to the opposite status
- Force override requires a reason (free text, required)
- Override is stored: warrantyStatus = forced value, warrantySource = forced, forceReason = admin's text
- Force reason visible to admin only. SC sees only the final In Warranty / Out of Warranty status.

Warranty state persistence across complaints

- Warranty status, expiry date, source, and force reason (if any) are saved on the Product record after every complaint.
- On the next complaint for the same product: warranty info is pre-shown from the Product record.
- If bill date was previously filled: auto-calculation runs fresh at new complaint time (today vs expiry date) — so a product that was In Warranty last year correctly shows Out of Warranty now if 3 years have passed.
- Admin can always update bill date or force override at any complaint creation or closing time.

Change 4 — Advanced Billing Filters + Mark as Paid System

Two improvements to the Billing Dashboard: richer filtering with exact date ranges, and a full payment tracking system for marking bills as paid or unpaid.

4A. Advanced Billing Filters

Current billing filters (SC, Month, Year, Product, Warranty) are replaced with a more flexible filter set:

Filter	Type	Details
Service Centre	Single select	All Service Centres OR a specific SC. Includes both registered and unregistered SCs.
From Date → To Date	Two date pickers	Exact calendar date range. Replaces the old Month + Year dropdowns. Admin picks any From and To date. Default on load: current month start to today.
Product	Single select	All Products / LED / Cooler
Warranty	Single select	All Warranty Status / In Warranty / Out of Warranty
Payment Status	Single select	All / Paid Only / Unpaid Only — NEW filter (see Section 4B)

- All filters work together simultaneously (AND logic).
- Reset Filters clears all back to defaults (All SCs, current month, all products, all warranty, all payment status).
- Both tabs — Per Complaint Bills and Monthly Invoice Summaries — respect all active filters.

4B. Mark as Paid System — Full Flow

Every closed bill now has a payment status: Unpaid (default on bill generation) or Paid. Admin can mark bills as paid at closing time or anytime later from the Billing Dashboard.

At Complaint Closing Time

- When admin confirms Done and the bill is generated, a checkbox/toggle appears: “Mark as Paid immediately” (optional).
- If admin checks it: bill is created with paymentStatus = paid, paidAt = current timestamp.
- If admin leaves it unchecked: bill is created with paymentStatus = unpaid. Can be marked paid later from Billing Dashboard.

From Billing Dashboard — 3 Ways to Mark as Paid

Method 1 — Bulk: All bills for selected SC + date range

- Admin selects a specific SC and a date range in filters.
- A “Mark All as Paid” button appears at the top of the filtered results.

- Clicking marks ALL currently visible unpaid bills in the filtered view as paid in one action.
- paidAt timestamp recorded as current time for all of them.

Method 2 — Bulk: All SCs in a date range

- Admin selects All Service Centres + a date range.
- Same “Mark All as Paid” button marks all unpaid bills across all SCs in that range.
- Useful for end-of-month batch payment processing.

Method 3 — Manual: Select individual bills

- Each bill row in the Per Complaint Bills table has a checkbox on the left.
- Admin selects any combination of individual bills (across any SC or date).
- A “Mark Selected as Paid” button appears in the action bar when at least one bill is selected.
- Only selected bills are marked paid. Others remain untouched.

Marking as Unpaid (Reversal)

- Any paid bill can be reversed back to unpaid.
- Same methods apply — bulk or individual selection.
- On clicking Mark as Unpaid: a warning is shown — “This will mark [N] paid bill(s) as unpaid. This action is reversible but will affect payment records. Continue?”
- Admin confirms → bills reverted to unpaid, paidAt cleared.
- Payment history is NOT stored per reversal — only the current paidAt timestamp is kept (last time it was marked paid).

Payment Record per Bill

Field	Stored On	Details
paymentStatus	Complaint / Invoice record	enum: unpaid / paid. Default: unpaid on bill generation.
paidAt	Complaint / Invoice record	Timestamp of when Mark as Paid was last pressed. Null if unpaid. Shown in billing detail view.
paidBy	Complaint / Invoice record	Which admin account marked it as paid. Stored for audit.

paidAt always reflects the LAST time the bill was marked paid. If a bill is marked paid, then unpaid, then paid again — paidAt shows the most recent paid timestamp only. Full reversal history is not stored.

Billing Table Display

Column / Element	Details
Payment Status column	Shows PAID (green badge) or UNPAID (amber badge) per row
Paid On column	Shows paidAt date+time if paid. Shows — if unpaid.

Column / Element	Details
Row checkbox	For manual individual selection
Action bar	Appears when any checkbox is selected — shows Mark Selected as Paid / Mark Selected as Unpaid buttons
Bulk action button	Mark All as Paid — shown above table, acts on current filtered view

4C. Running Totals — Two Sums Below Billing Table

Below the Per Complaint Bills table, two running totals are always shown based on the current active filters and date range:

Total	What It Shows
Total (all bills in view)	Sum of ALL bills currently shown (paid + unpaid combined) for the selected filters and date range.
Unpaid Total	Sum of only UNPAID bills in the current view. This is what Microvison still owes the SC(s) in the selected period.

- Both totals update live as filters change.
- If a specific SC is selected: totals reflect that SC only.
- If All Service Centres is selected: totals reflect all SCs combined for the selected date range.
- Default view on opening Billing Dashboard (no filters changed): shows current month, all SCs — totals give instant overview of this month's billing situation.

Master Summary — All Changes So Far

Change	What It Does
1A — Skip SC Assignment	Step 5 (was Step 4) can be skipped. Complaint created as unassigned. Admin assigns later from detail view. WA-01 fires on that later assignment.
1B — Advanced Search	Top bar searches 5 fields simultaneously. All filters combinable. Date shortcuts: Today, Yesterday, Last 7, Last 30, Custom.
2A — Unregistered SC creation	Admin creates minimal SC record on spot during assignment. isUnregistered flag. No portal, no WP reminders to SC.
2B — WA-04 still fires	Customer gets WhatsApp even for unregistered SC assignment. All other SC-related WP messages do not fire.
2C — Admin maintains updates	Admin fills all forms, updates all statuses for unregistered SC on their behalf. Full billing and product tracking still works.
2D — New city/district/state	Admin creates new locations on the spot from any city field. Instantly available system-wide.
2E — Searchable and reusable	Unregistered SCs appear in SC list, are filterable, searchable, and reusable for future complaints.

Change	What It Does
2F — Upgrade to registered	SC submits normal registration. Admin links via manual search (not phone-match only). All history transfers. Badge removed.
3 — New Step 2	New Step 2 inserted: Bill Date, Bill Photo, Shop Name, Serial Number, Model Number. All optional but recommended. Old steps shift by 1.
3C — Location / Action Text	Optional field in Step 1. Admin pastes Maps link or location note. Sent to SC in WA-01.
3D — SC Done demands	Missing bill info → SC must upload bill photo. Missing serial/model → SC must upload serial sticker photo. Both bypassable with warning.
3E — SC uploads auto-fill product	SC-uploaded bill photo auto-saves to Product record. Admin can keep, remove, or replace. Serial slip stored separately for admin to transcribe.
3F — Admin closing check	Before closing, admin warned of missing Step 2 fields. Can fill or bypass per field. Persistent warning on Product record for future complaints.
3G — Warranty logic	Bill date + 3yr auto-calc. No bill date → manual selection + reason. LED installation defaults to In Warranty. Force override always available with reason. Reason visible to admin only.
4A — Advanced billing filters	Exact date range (From → To) replaces Month + Year. SC, Product, Warranty filters remain. New Payment Status filter: All / Paid Only / Unpaid Only. All combinable simultaneously.
4B — Mark as Paid system	Bills default to unpaid on generation. Mark paid at closing (optional) or from Billing Dashboard — bulk all in filtered view, bulk all SCs in date range, or manual individual checkbox selection. Reversal to unpaid allowed with warning. paidAt + paidBy stored per bill. paidAt = last time marked paid.
4C — Running totals	Two live sums below billing table: Total (all bills in view) and Unpaid Total (only unpaid). Both update with active filters. Default = current month, all SCs.

End of Current Changes · More changes will be added to this document as discussed